

## 1. Describe a time when you led a cross-functional team to deliver a complex project.

### **S – Situation:**

At MathWorks, we were tasked with modernizing MATLAB's legacy Data Exploration and App Building workflows, which were fragmented and had performance bottlenecks across multiple web domains.

### **T – Task:**

As a Senior Front-End Developer, I was responsible for leading a cross-functional team of UI engineers, backend developers, UX designers, and QA testers to revamp and unify these workflows for the next generation of MATLAB web apps.

### **A – Action:**

I initiated Agile processes—sprint planning, retrospectives, and daily stand-ups—using JIRA. I designed a component-based architecture using ReactJS and JavaScript UI design patterns to ensure maintainability and scalability. I created shared documentation, mentored junior engineers, and facilitated stakeholder meetings to align technical deliverables with business goals.

### **R – Result:**

The project significantly improved the product's responsiveness and user satisfaction. Our modular architecture reduced onboarding time and was adopted across multiple internal teams. The project was recognized in internal showcases and served as a benchmark for future platform modernization efforts.

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## 2. Tell me about a time when you significantly improved the UI/UX of an application.

### **S – Situation:**

At MathWorks, multiple web-based domains of MATLAB had evolved independently, leading to inconsistencies in design, navigation, and user interaction.

### **T – Task:**

My task was to create a unified UI/UX framework that standardized user experience across all platforms while incorporating accessibility and responsive design.

### **A – Action:**

I collaborated with UX designers and product owners to identify friction points. I built a modular, component-based architecture in JavaScript and ReactJS, introduced accessibility standards (ARIA roles, keyboard navigation), and ensured responsiveness across devices. We conducted internal and external usability testing to iterate on the design.

### **R – Result:**

The unified framework led to a more cohesive user experience and reduced confusion. Support tickets related to UI dropped by 25%, and the framework became the standard for all future MATLAB web applications.

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## 3. Give an example of how you mentored a junior developer or onboarded a new team member.

### **S – Situation:**

A junior developer joined our team with limited experience in React and modern frontend best practices.

**T – Task:**

I was responsible for onboarding the developer and accelerating their transition to becoming an independent contributor.

**A – Action:**

I created an onboarding guide covering our UI framework, coding standards, and test strategies. I held regular pair programming sessions, introduced them to Agile workflows, and reviewed their code with constructive feedback. I also helped them understand the business value behind their tasks.

**R – Result:**

The developer was able to independently deliver a new UI component within three months and presented it during a team demo. My onboarding structure was later formalized and reused for new team hires.

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**4. Describe a time when you worked across multiple teams to complete a project.****S – Situation:**

We needed to build shared UI components that would be reused by both cloud and desktop product teams at MathWorks.

**T – Task:**

I was responsible for designing reusable components and facilitating integration with several teams that had different priorities and release schedules.

**A – Action:**

I organized planning sessions to align requirements and delivery timelines. I created integration guides and hosted a central Confluence space. I also set up a Slack channel for real-time feedback and issue resolution, and aligned everyone on shared code quality standards and testing coverage.

**R – Result:**

Our components were successfully adopted by three teams within two releases. The clear documentation and proactive support reduced integration issues and made the approach a model for future cross-team work.

**5. Tell me about a time you improved testing processes for better quality assurance.****S – Situation:**

The legacy UI testing infrastructure was not sufficient for the scale and complexity of new React-based features, leading to missed bugs in production.

**T – Task:**

I needed to revamp the testing approach to support scalable, automated, and meaningful test coverage.

**A – Action:**

I implemented a new stack using QUnit for unit testing, SinonJS for mocks, and Selenium for end-to-end tests. I integrated these into our CI pipeline and set up code coverage tools. I also conducted training sessions on writing effective tests and initiated shared regression test suites with QA.

**R – Result:**

Code coverage improved by 40%, and post-release defects were significantly reduced. The new approach became the template for other UI teams at MathWorks.

## **6. Tell me about a time you received constructive criticism and how you handled it.**

### **S – Situation:**

During a feature sprint, I received feedback that a component I built didn't align with the updated UX design system.

### **T – Task:**

I had to respond professionally, fix the issues, and prevent such mismatches in the future.

### **A – Action:**

I reached out to the designer for a 1-on-1 session to better understand the expectations. I refactored the component to align with the latest specs. To improve collaboration, I proposed and established a design-dev review checkpoint and updated our internal UI checklist.

### **R – Result:**

The redesigned component met all UX criteria, and our new process reduced future design misalignments. It also improved our cross-functional trust and efficiency.

## **7. Share a time you worked under a tight deadline.**

### **S – Situation:**

Ahead of a major MATLAB user conference, we had to demo a new UI toolkit within four weeks.

### **T – Task:**

I had to deliver all required components within this strict timeline while maintaining quality.

### **A – Action:**

I decomposed tasks using MoSCoW prioritization and tracked delivery with JIRA. I led concurrent QA and development cycles and worked extended hours when needed. I also ensured daily syncs with stakeholders to monitor progress and unblock issues quickly.

### **R – Result:**

The components were delivered on time, and the live demo received excellent feedback. The project was praised in our sprint retrospective for being one of the most efficiently executed ones.

## **8. Describe how you handled a major tech migration.**

### **S – Situation:**

Our team was tasked with moving away from the Dojo/Dijit framework to modern ReactJS-based architecture for future UI development.

### **T – Task:**

I led the migration initiative while ensuring legacy systems remained functional and compatible.

### **A – Action:**

I evaluated frameworks, built POCs, and advocated for ReactJS. I trained the team via internal workshops and created React wrappers for legacy components to ensure backward compatibility. I also ran feature parity checks and defined a phased migration roadmap.

### **R – Result:**

The migration happened smoothly with zero downtime. React enabled faster development, better testing, and consistent UI practices. Our team's approach became a reference for future migration efforts at MathWorks.

## **9. Tell me about a process improvement you introduced.**

### **S – Situation:**

We lacked consistent unit test practices, resulting in unreliable test coverage and production issues.

### **T – Task:**

I wanted to introduce a culture of TDD and enforce quality gates via CI.

### **A – Action:**

I implemented Jest, integrated code coverage metrics in our CI pipeline, and enforced minimum coverage thresholds for PRs. I also created boilerplate test templates and hosted weekly test training sessions.

### **R – Result:**

Test coverage grew from 35% to 75% in one quarter. Developers gained confidence to refactor and deploy quickly without breaking things. The change reduced production bugs significantly.

## **10. Share an example of working closely with stakeholders to ensure product success.**

### **S – Situation:**

During our modular UI framework rollout, different teams had diverse and evolving needs, which often conflicted.

### **T – Task:**

My goal was to unify stakeholder expectations and build a framework that could adapt to team-specific customization needs without compromising architectural integrity.

### **A – Action:**

I led bi-weekly stakeholder meetings and created shared requirement docs on Confluence. I tracked feedback and trade-offs transparently and became the go-to point of contact for decisions and clarifications.

### **R – Result:**

The framework was adopted by six teams within the first quarter. The structured feedback loop helped resolve conflicts early, and stakeholder trust grew as a result of the transparent and inclusive development approach.